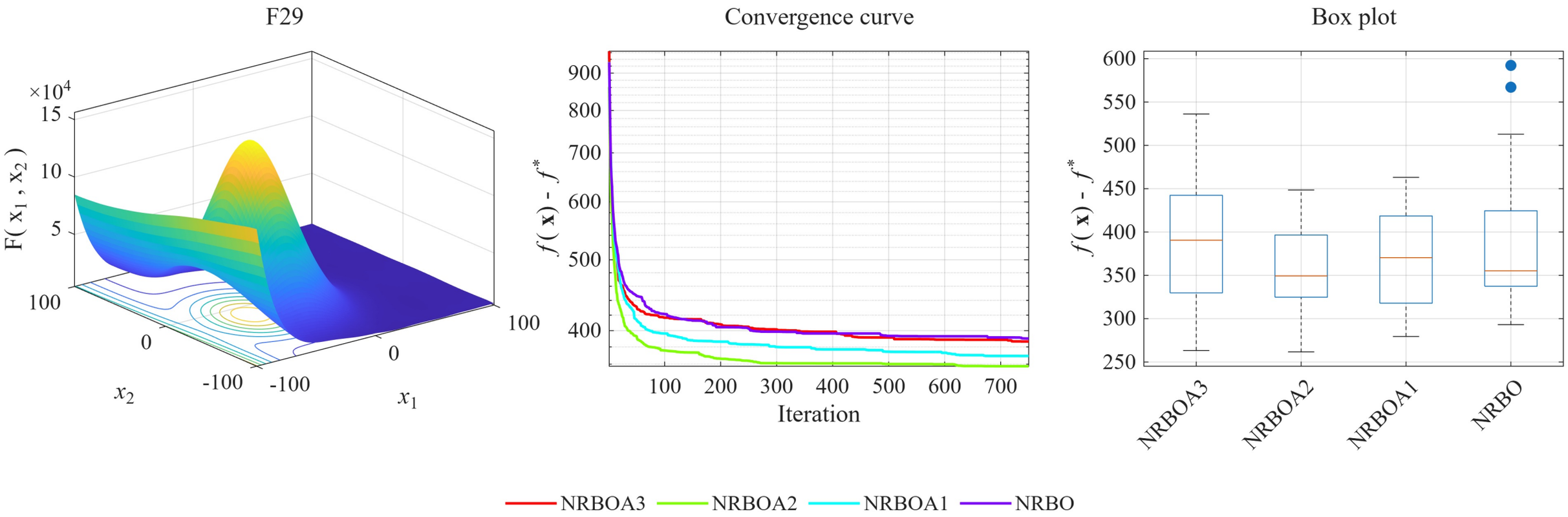

F29
×10^4
15
10
5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
900
800
700
600
500
400
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
600
550
500
450
400
350
300
250
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO